Drew University - CSCI 260 - Fall 2024
In-Class Exercise 5 - C Pointers

Instruction: Please work in groups of 4 and write down your names for attendance. 12/02/2024
Names:

1. Without running the following program, identify its output. If line 4 is "int x = 7" instead, what is the output then?

```
int x;              // I am a global variable
int main() {
    int *px = &x;
    x = 7;
    *px = 4;
    printf("%d\n", x);
    return 0;
}
```

2. Without running the following program, identify its output.

```
void func(int* ptr) {
    *ptr = 30;
}
int main() {
    int val = 10;
    func(&val);
    printf("%d\n", val);
    return 0;
}
```

3. Without running the following program, identify its output.

```
int main() {
    int x;
    int *ptr = &x;
    *ptr = 5;
    printf("x = %d, *ptr = %d\n", x, *ptr);
    *ptr += 3;
    printf("x = %d, *ptr = %d\n", x, *ptr);
    (*ptr)++;
    printf("x = %d, *ptr = %d\n", x, *ptr);
    return 0;
}
```

4. The following function of converting letters in a string to lower-cases has some issues. In fact, if you try to print the returned string, nothing will be printed out. The issue is with the return type not being a char*. Run the program and try to fix it.

```
#define MAX_LEN 20
char* toLowerCase(char* str) {
    char newStr[MAX_LEN];
    int i;
    for (i = 0; i < MAX_LEN; i++) {
        if ('A' <= str[i] && str[i] <= 'Z')
            newStr[i] = str[i] + ('a' - 'A');
        else
            newStr[i] = str[i];
    }
    return newStr;
}
```

5. Recall when swapping 2 numbers using a *swap_ptr()* function, we passed the addresses of the numbers as arguments and switched what the pointers were pointing towards in the function.

   Although it seems that the following *swap_string()* function for swapping 2 strings utilizes the same mechanism as *swap_ptr()*, it is NOT working as intended.

   Run the program and try to identify the problem (i.e., what makes the passing in this function different from that in *swap_ptr()?*) and then fix it.

```
void swap_strings(char* s1, char* s2) {
    char* temp = s1;
    s1 = s2;
    s2 = temp;
}
int main() {
    char *s1 = "Hello";
    char *s2 = "World";
    swap_strings(s1, s2);
    return 0;
}
```